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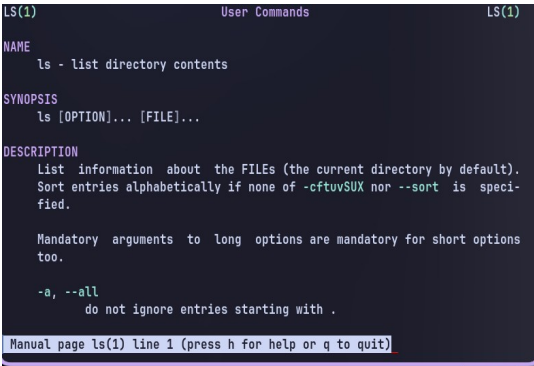
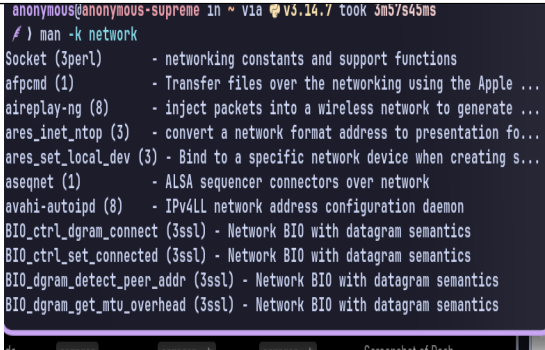
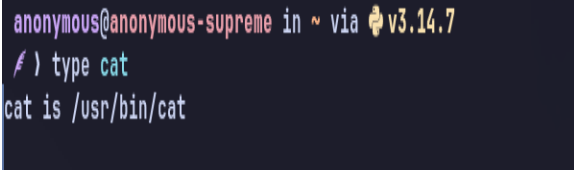
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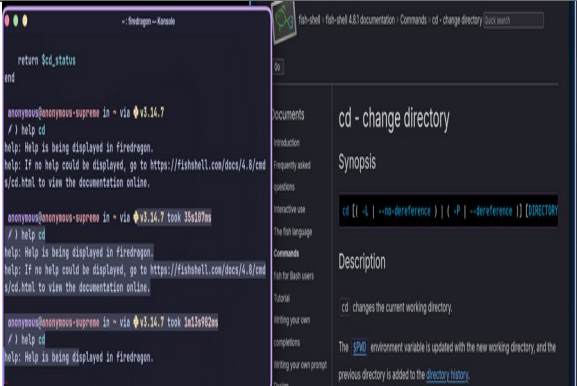
LOS_Lab Assignment-2

The main aim of this lab session is to provide hands-on experience on

- Getting Help**
- Basic Commands**
- Navigation**
- File System**
- Simple shell script**
- Globbing**
- Extended Globbing**
- Filters -I (head,tail,wc)**

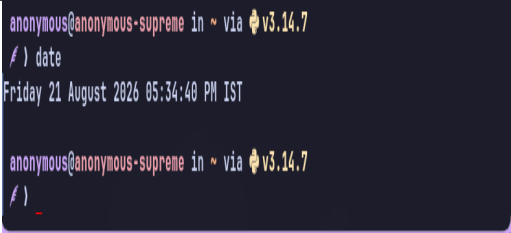
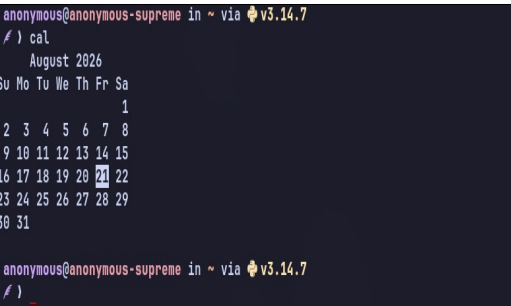
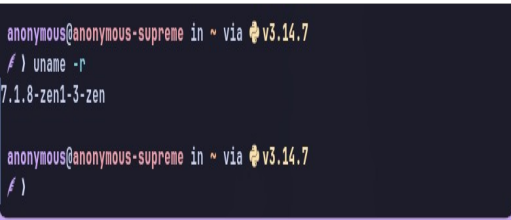
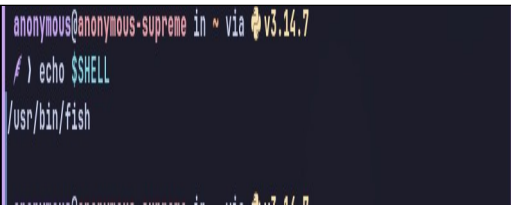
1. Getting Help

Task	Command Name	Syntax	Example	Screenshots
To get manual page for the known command	man	man <command>	man ls	 <pre> LS(1) User Commands LS(1) NAME ls - list directory contents SYNOPSIS ls [OPTION]... [FILE]... DESCRIPTION List information about the FILES (the current directory by default). Sort entries alphabetically if none of -cftuvSUX nor --sort is speci- fied. Mandatory arguments to long options are mandatory for short options too. -a, --all do not ignore entries starting with . Manual page ls(1) line 1 (press h for help or q to quit) </pre>
To get manual page for the unknown command	man -k / apropos	man -k <keyword>	man -k network	 <pre> anonymous@anonymous-supreme in ~ via v3.14.7 took 3ms/545ms #) man -k network Socket (3perl) - networking constants and support functions afpcmd (1) - Transfer files over the networking using the Apple ... aireplay-ng (8) - inject packets into a wireless network to generate ... ares_inet_ntop (3) - convert a network format address to presentation fo... ares_set_local_dev (3) - Bind to a specific network device when creating s... aseqnet (1) - ALSA sequencer connectors over network avahi-autoipd (8) - IPv4LL network address configuration daemon BIO_ctrl_dgram_connect (3ssl) - Network BIO with datagram semantics BIO_ctrl_set_connected (3ssl) - Network BIO with datagram semantics BIO_dgram_detect_peer_addr (3ssl) - Network BIO with datagram semantics BIO_dgram_get_mtu_overhead (3ssl) - Network BIO with datagram semantics </pre>
To know the source file binary	type	type <command>	type ls/cat	 <pre> anonymous@anonymous-supreme in ~ via v3.14.7 #) type cat cat is /usr/bin/cat </pre>

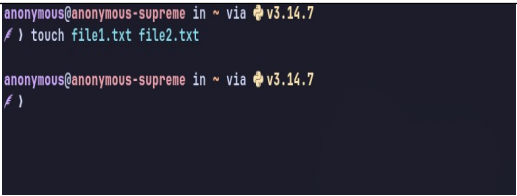
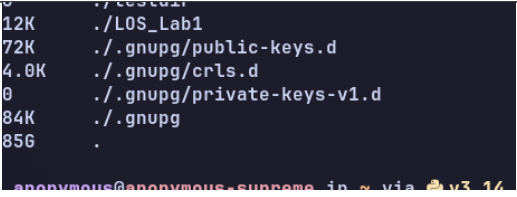
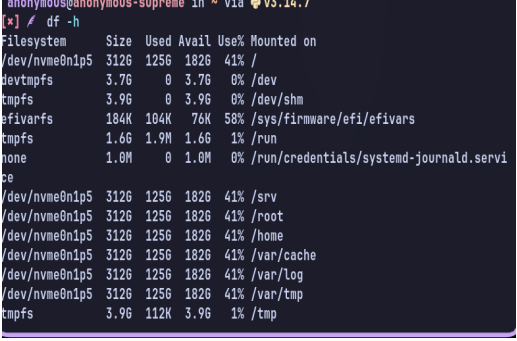
To know the path of the command	which	which <command>	which ls	<pre> anonymous@anonymous-supreme in ~ via 🍀 v3.14.7 took 1m35s640ms [🍀] / which ls /usr/bin/ls anonymous@anonymous-supreme in ~ via 🍀 v3.14.7 </pre>
To know the command is external or internal	type	type <command>	type cd	<pre> anonymous@anonymous-supreme in ~ via 🍀 v3.14.7 /) type cd cd is a function with definition Defined in embedded:functions/cd.fish @ line 5 function cd --description 'Change directory' set -l MAX_DIR_HIST 25 # Skip history in subshells. if status is-command-substitution builtin cd \$argv end end </pre>
To get help for the internal command	help / man	help <command>	help cd	

To list out bash commands	builtin	builtin -n	builtin -n	<pre> anonymous@anonymous-supreme in ~ via 📶 v3.14.7 took 1m49s998ms [~] / builtin -n ! : [- abbr and argparse begin bg bind block break breakpoint builtin </pre>
To know the usage of the command	--help	<command> --help	ls --help	<pre> anonymous@anonymous-supreme in ~ via 📶 v3.14.7 *) / ls --help modern replacement for ls Usage: eza [OPTIONS] [FILE]... META OPTIONS: --stdin read file names from stdin -?, --help Print help -v, --version Print help LAYOUT OPTIONS: -1, --oneline display one entry per line -l, --long display extended file metadata as a table -G, --grid display entries as a grid (default) -x, --across sort the grid across, rather than downwards -R, --recurse recurse into directories -T, --tree recurse into directories as a tree </pre>

2. Basic Commands

Task	Command Name	Syntax	Example	Screenshots
To know today's date	date	date	date	 <pre>anonymous@anonymous-supreme in ~ via v3.14.7 #) date Friday 21 August 2026 05:34:40 PM IST anonymous@anonymous-supreme in ~ via v3.14.7 #) _</pre>
To print calendar	cal	cal	cal	 <pre>anonymous@anonymous-supreme in ~ via v3.14.7 #) cal August 2026 Su Mo Tu We Th Fr Sa 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 anonymous@anonymous-supreme in ~ via v3.14.7 #) _</pre>
To print kernel version	uname	uname -r	uname -r	 <pre>anonymous@anonymous-supreme in ~ via v3.14.7 #) uname -r 7.1.8-zen1-3-zen anonymous@anonymous-supreme in ~ via v3.14.7 #) _</pre>
To print default shell	echo	echo \$SHELL	echo \$SHELL	 <pre>anonymous@anonymous-supreme in ~ via v3.14.7 #) echo \$SHELL /usr/bin/fish anonymous@anonymous-supreme in ~ via v3.14.7 #) _</pre>

To print currently logged in user	whoami	whoami	whoami	<pre> anonymous@anonymous-supreme in ~ via v3.14.7 #) whoami anonymous anonymous@anonymous-supreme in ~ via v3.14.7 #) _ </pre>
To create shortcut for command	alias	alias name='command'	alias ll='ls -la'	<pre> anonymous@anonymous-supreme in ~ via v3.14.7 #) alias ll='ls -la' anonymous@anonymous-supreme in ~ via v3.14.7 #) _ </pre>
To delete shortcut	alias	functions -e <name>	functions -e ll	<pre> anonymous@anonymous-supreme in ~ via v3.14.7 took 3s573ms CQ) functions -e ll anonymous@anonymous-supreme in ~ via v3.14.7 #) alias ll alias: body cannot be empty anonymous@anonymous-supreme in ~ via v3.14.7 *) _ </pre>
To change the timestamp of the file	touch	touch <filename>	touch file.txt	<pre> anonymous@anonymous-supreme in ~ via v3.14.7 [*] touch file.txt anonymous@anonymous-supreme in ~ via v3.14.7 #) _ </pre>

To clear the screen	clear	clear	clear	
To create empty files	touch	touch <filename>	touch file1.txt file2.txt	
To know disk usage	du	du -h	du -h	
To know free space in the system	df	df -h	df -h	

To know about the Linux release	lsb_release	lsb_release -a	lsb_release -a	<pre> anonymous@anonymous-supreme in ~ via 🍀 v3.14.7 /) lsb_release -a SB Version: n/a istributor ID: Garuda escription: Garuda Linux Broadwing elease: Soaring odename: Broadwing anonymous@anonymous-supreme in ~ via 🍀 v3.14.7 </pre>
To show login history	last	last	last	<pre> anonymo* pts/0 Thu Mar 27 17:37 - 17:40 (00:00) anonymo* tty1 Thu Mar 27 17:38 - down (00:44) reboot system boot 6.13.8-zen1-1-z* Thu Mar 27 17:38 - 18:22 (00:44) anonymo* pts/0 :1 Thu Mar 27 17:36 - 17:37 (00:00) anonymo* pts/0 :1 Thu Mar 27 17:26 - 17:33 (00:06) anonymo* pts/0 :1 Thu Mar 27 17:17 - 17:25 (00:08) anonymo* tty1 Thu Mar 27 17:14 - down (00:23) reboot system boot 6.13.5-zen1-1-z* Thu Mar 27 17:14 - 17:37 (00:23) wtmp begins Thu Mar 27 17:14:16 2025 anonymous@anonymous-supreme in ~ via 🍀 v3.14.7 /) </pre>
To check the current user privileges	sudo	sudo -l	sudo -l	<pre> anonymous@anonymous-supreme in ~ via 🍀 v3.14.7 /) sudo -l [sudo] password for anonymous: Matching Defaults entries for anonymous on anonymous-supreme: secure_paths=/usr/local/sbin:/usr/local/bin:/usr/bin, editor=/usr/bin/micro, insults, pwfeedback Runas and Command-specific defaults for anonymous: Defaults!/usr/bin/visudo env_keep+="SUDO_EDITOR EDITOR VISUAL" User anonymous may run the following commands on anonymous-supreme: (ALL) ALL anonymous@anonymous-supreme in ~ via 🍀 v3.14.7 as 🍀 took 55346ms /) </pre>
To find about the Linux release	cat	cat /etc/os-release	cat /etc/os-release	<pre> anonymous@anonymous-supreme in ~ via 🍀 v3.14.7 as 🍀 took 55346ms /) cat /etc/os-release NAME="Garuda Linux" PRETTY_NAME="Garuda Linux" ID=garuda ID_LIKE=arch VARIANT_ID=rolling NSI_COLOR="38;2;23;147;209" HOME_URL="https://garudalinux.org/" DOCUMENTATION_URL="https://wiki.garudalinux.org/" SUPPORT_URL="https://forum.garudalinux.org/" BUG_REPORT_URL="https://gitlab.com/groups/garuda-linux/" PRIVACY_POLICY_URL="https://terms.archlinux.org/docs/privacy-policy/" LOGO=garudalinux anonymous@anonymous-supreme in ~ via 🍀 v3.14.7 as 🍀 /) </pre>

To check system uptime	uptime	uptime	uptime	<pre> anonymous@anonymous-supreme in ~ via v3.14.7 as /) uptime 17:55:44 up 3:41, 1 user, load average: 0.83, 0.83, 0.85 anonymous@anonymous-supreme in ~ via v3.14.7 as /) _ </pre>
To shutdown	shutdown	sudo shutdown now	sudo shutdown now	<pre> anonymous@anonymous-supreme in ~ via v3.14.7 as /) sudo shutdown now </pre>
To restart	reboot	sudo reboot	sudo reboot	<pre> anonymous@anonymous-supreme in ~ via v3.14.7 as /) sudo reboot_ </pre>

3. Navigation

Task	Command	Syntax	Screenshots
To navigate home directory	cd	cd	<pre> anonymous@anonymous-supreme in ~ via v3.14.7 as /) cd anonymous@anonymous-supreme in ~ via v3.14.7 </pre>
To navigate to the parent directory	cd ..	cd ..	<pre> anonymous@anonymous-supreme in ~ via v3.14.7 /) cd .. anonymous@anonymous-supreme in /home /) cd .. </pre>
To navigate to the child directory	cd <directory>	cd <directory>	<pre> anonymous@anonymous-supreme in ~ via v3.14.7 /) cd Downloads anonymous@anonymous-supreme in ~/Downloads via C v16.2.1-gcc /) </pre>

Alternate command to cd	pushd	pushd <directory>	<pre> anonymous@anonymous-supreme in ~ via 🍌 v3.14.7 /) pushd Downloads anonymous@anonymous-supreme in ~/Downloads via C v16.2.1-gcc /) _ </pre>
To go back to the previous directory	cd -	cd -	<pre> anonymous@anonymous-supreme in ~/Downloads via C v16.2.1-gcc /) cd ~ anonymous@anonymous-supreme in ~ via 🍌 v3.14.7 /) </pre>
To go to the root directory	cd /	cd /	<pre> anonymous@anonymous-supreme in ~ via 🍌 v3.14.7 /) cd / anonymous@anonymous-supreme in / /) </pre>

4. File System

Task	Syntax	Command
How to identify the file system	df -T <path>	<pre> anonymous@anonymous-supreme in ~ via 🍌 v3.14.7 /) df -T / Filesystem Type 1K-blocks Used Available Use% Mounted on /dev/nvme0n1p5 btrfs 326281700 130840204 190507012 41% / anonymous@anonymous-supreme in ~ via 🍌 v3.14.7 </pre>

5. Globbing

Create a dedicated practice directory containing sample files such as:

index.html, login.html, admin.htm, report.txt, access.log, error.log, user1.txt, user2.log, file123, backup1.html, backup2.txt

```
anonymous@anonymous-supreme in ~ via v3.14.7
# ) mkdir -p ~/globbing_practice

anonymous@anonymous-supreme in ~ via v3.14.7
# ) cd ~/globbing_practice

anonymous@anonymous-supreme in ~/globbing_practice
# ) touch index.html login.html admin.htm report.txt access.log error.log user1.txt user2.log file123 backup1.html backup2.txt

anonymous@anonymous-supreme in ~/globbing_practice
# ) ls
.rw-r--r-- 0 anonymous 21 Aug 18:27 access.log
.rw-r--r-- 0 anonymous 21 Aug 18:27 admin.htm
.rw-r--r-- 0 anonymous 21 Aug 18:27 backup1.html
.rw-r--r-- 0 anonymous 21 Aug 18:27 backup2.txt
.rw-r--r-- 0 anonymous 21 Aug 18:27 error.log
.rw-r--r-- 0 anonymous 21 Aug 18:27 file123
.rw-r--r-- 0 anonymous 21 Aug 18:27 index.html
.rw-r--r-- 0 anonymous 21 Aug 18:27 login.html
.rw-r--r-- 0 anonymous 21 Aug 18:27 report.txt
.rw-r--r-- 0 anonymous 21 Aug 18:27 user1.txt
.rw-r--r-- 0 anonymous 21 Aug 18:27 user2.log

anonymous@anonymous-supreme in ~/globbing_practice
# )
```

Use **globbing patterns wherever possible**, rather than specifying filenames individually.

- List all files ending in .html.
- List all files **other than .html files** using extended globbing.
- List filenames containing at least one numeric digit.
- List files whose names begin with user followed by a single digit.
- List files ending in either .txt or .log using a single **extended glob pattern**.
- List files that do **not** end in .txt or .log.
- List files beginning with either access or error using extended globbing.
- List .html files whose filenames contain a numeric digit.

```
➤ ) ls *.html
-rw-r--r-- 0 anonymous 21 Aug 18:27 backup1.html
-rw-r--r-- 0 anonymous 21 Aug 18:27 index.html
-rw-r--r-- 0 anonymous 21 Aug 18:27 login.html

anonymous@anonymous-supreme in ~/globbing_practice
➤ ) ls !(*.html)
-rw-r--r-- 0 anonymous 21 Aug 18:27 access.log
-rw-r--r-- 0 anonymous 21 Aug 18:27 admin.htm
-rw-r--r-- 0 anonymous 21 Aug 18:27 backup2.txt
-rw-r--r-- 0 anonymous 21 Aug 18:27 error.log
-rw-r--r-- 0 anonymous 21 Aug 18:27 file123
-rw-r--r-- 0 anonymous 21 Aug 18:27 report.txt
-rw-r--r-- 0 anonymous 21 Aug 18:27 user1.txt
-rw-r--r-- 0 anonymous 21 Aug 18:27 user2.log

anonymous@anonymous-supreme in ~/globbing_practice
➤ ) ls *[0-9]*
-rw-r--r-- 0 anonymous 21 Aug 18:27 backup1.html
-rw-r--r-- 0 anonymous 21 Aug 18:27 backup2.txt
-rw-r--r-- 0 anonymous 21 Aug 18:27 file123
-rw-r--r-- 0 anonymous 21 Aug 18:27 user1.txt
-rw-r--r-- 0 anonymous 21 Aug 18:27 user2.log

anonymous@anonymous-supreme in ~/globbing_practice
➤ ) ls user[0-9].*
-rw-r--r-- 0 anonymous 21 Aug 18:27 user1.txt
-rw-r--r-- 0 anonymous 21 Aug 18:27 user2.log

anonymous@anonymous-supreme in ~/globbing_practice
➤ ) ls *.{txt,log}
-rw-r--r-- 0 anonymous 21 Aug 18:27 access.log
-rw-r--r-- 0 anonymous 21 Aug 18:27 backup2.txt
-rw-r--r-- 0 anonymous 21 Aug 18:27 error.log
-rw-r--r-- 0 anonymous 21 Aug 18:27 report.txt
-rw-r--r-- 0 anonymous 21 Aug 18:27 user1.txt
-rw-r--r-- 0 anonymous 21 Aug 18:27 user2.log

anonymous@anonymous-supreme in ~/globbing_practice
➤ ) ls !(*.txt|*.log)
-rw-r--r-- 0 anonymous 21 Aug 18:27 admin.htm
-rw-r--r-- 0 anonymous 21 Aug 18:27 backup1.html
-rw-r--r-- 0 anonymous 21 Aug 18:27 file123
-rw-r--r-- 0 anonymous 21 Aug 18:27 index.html
-rw-r--r-- 0 anonymous 21 Aug 18:27 login.html

anonymous@anonymous-supreme in ~/globbing_practice
➤ ) ls @(access|error)*
-rw-r--r-- 0 anonymous 21 Aug 18:27 access.log
-rw-r--r-- 0 anonymous 21 Aug 18:27 error.log

anonymous@anonymous-supreme in ~/globbing_practice
➤ ) ls *[0-9]*.html
-rw-r--r-- 0 anonymous 21 Aug 18:27 backup1.html

anonymous@anonymous-supreme in ~/globbing_practice
➤ )
```

6. Globbing

a. Assume a directory contains:

access.log, error.log, auth.log, auth.log.1, auth.log.2, system.log, notes.txt

Write an extended glob pattern that selects only access.log, error.log, and auth.log.

Select all files except notes.txt.

Select files named either auth.log.1 or auth.log.2 using pattern matching rather than writing both complete filenames separately.

b. Given the files:

malware1.log, malware2.log, malwareA.log, malware_report.txt, normal.log

construct a glob pattern that identifies .log files whose names start with malware and contain a numeric character.

c. Given:

failed_login.log, successful_login.log, login_backup.log, login.txt, firewall.log

use extended globbing to select .log files containing either failed_login or successful_login.

Construct a pattern that lists all .log files **except** firewall.log.

```
anonymous@anonymous-supreme in ~/globbing_bash
# ) ls @(access|error|auth).log
.rw-r--r-- 0 anonymous 21 Aug 19:00 access.log
.rw-r--r-- 0 anonymous 21 Aug 19:00 auth.log
.rw-r--r-- 0 anonymous 21 Aug 19:00 error.log

anonymous@anonymous-supreme in ~/globbing_bash
# ) ls !(notes.txt)
.rw-r--r-- 0 anonymous 21 Aug 19:00 access.log
.rw-r--r-- 0 anonymous 21 Aug 19:00 auth.log
.rw-r--r-- 0 anonymous 21 Aug 19:00 auth.log.1
.rw-r--r-- 0 anonymous 21 Aug 19:00 auth.log.2
.rw-r--r-- 0 anonymous 21 Aug 19:00 error.log
.rw-r--r-- 0 anonymous 21 Aug 19:00 failed_login.log
.rw-r--r-- 0 anonymous 21 Aug 19:00 firewall.log
.rw-r--r-- 0 anonymous 21 Aug 19:00 login.txt
.rw-r--r-- 0 anonymous 21 Aug 19:00 login_backup.log
.rw-r--r-- 0 anonymous 21 Aug 19:00 malware1.log
.rw-r--r-- 0 anonymous 21 Aug 19:00 malware2.log
.rw-r--r-- 0 anonymous 21 Aug 19:00 malware_report.txt
.rw-r--r-- 0 anonymous 21 Aug 19:00 malwareA.log
.rw-r--r-- 0 anonymous 21 Aug 19:00 normal.log
.rw-r--r-- 0 anonymous 21 Aug 19:00 successful_login.log
.rw-r--r-- 0 anonymous 21 Aug 19:00 system.log

anonymous@anonymous-supreme in ~/globbing_bash
# ) ls auth.log.[12]
.rw-r--r-- 0 anonymous 21 Aug 19:00 auth.log.1
.rw-r--r-- 0 anonymous 21 Aug 19:00 auth.log.2

anonymous@anonymous-supreme in ~/globbing_bash
# ) ls malware[0-9]*.log
.rw-r--r-- 0 anonymous 21 Aug 19:00 malware1.log
.rw-r--r-- 0 anonymous 21 Aug 19:00 malware2.log

anonymous@anonymous-supreme in ~/globbing_bash
# ) ls @(failed_login|successful_login).log
.rw-r--r-- 0 anonymous 21 Aug 19:00 failed_login.log
.rw-r--r-- 0 anonymous 21 Aug 19:00 successful_login.log

anonymous@anonymous-supreme in ~/globbing_bash
# ) ls !(*.log|firewall.log)
.rw-r--r-- 0 anonymous 21 Aug 19:00 auth.log.1
.rw-r--r-- 0 anonymous 21 Aug 19:00 auth.log.2
.rw-r--r-- 0 anonymous 21 Aug 19:00 login.txt
.rw-r--r-- 0 anonymous 21 Aug 19:00 malware_report.txt
.rw-r--r-- 0 anonymous 21 Aug 19:00 notes.txt

anonymous@anonymous-supreme in ~/globbing_bash
# )
```

7.Filters — Complete Command Table

1. First Entries — head

No.	Task	Command
1	Display the first 10 entries of login.log	head -n 10 login.log
2	Display the first 5 entries of access.log	head -n 5 access.log
3	Display the first 15 security events from security.log	head -n 15 security.log
4	Display the first 20 events recorded in incident.log	head -n 20 incident.log
5	Display only the first entry in login.log	head -n 1 login.log
6	Display the first 25 web requests recorded in access.log	head -n 25 access.log

2. Last Entries — tail

No.	Task	Command
7	Display the last 10 entries of login.log	tail -n 10 login.log
8	Display the last 5 entries of access.log	tail -n 5 access.log
9	Display the 20 most recent security events from security.log	tail -n 20 security.log
10	Display the 25 most recent events from incident.log	tail -n 25 incident.log
11	Display only the last entry in security.log	tail -n 1 security.log
12	Display the last 50 events recorded in incident.log	tail -n 50 incident.log

3. Starting From a Particular Line

No. .	Task	Command
13	Display login.log starting from line 41 until the end	tail -n +41 login.log
14	Display security.log starting from line 101 until the end	tail -n +101 security.log

4. Counting — wc

No.	Task	Command
15	Find the total number of lines in login.log	wc -l login.log
16	Find the total number of lines in access.log	wc -l access.log
17	Find the total number of security events in security.log	wc -l security.log
18	Determine the total number of events in incident.log	wc -l incident.log
19	Count the number of words in login.log	wc -w login.log
20	Count the number of bytes in security.log	wc -c security.log
21	Display the line, word, and byte counts of access.log	wc -l -w -c access.log
22	Find line counts of all four .log files using one wc command	wc -l login.log access.log security.log incident.log

5. Combining head and tail Using Pipes

No.	Task	Command
23	First 20 entries of login.log, then display the last 5	head -n 20 login.log tail -n 5
24	Last 20 entries of login.log, then display the first 5	tail -n 20 login.log head -n 5
25	First 30 entries of access.log, then display the last 10	head -n 30 access.log tail -n 10
26	Last 30 entries of access.log, then display the first 10	tail -n 30 access.log head -n 10
27	First 50 events of security.log, then display the last 10	head -n 50 security.log tail -n 10
28	Last 50 events of security.log, then display the first 10	tail -n 50 security.log head -n 10

6. Extracting Specific Line Ranges

No.	Task	Command
29	Display entries from line 11 to line 20 of login.log	head -n 20 login.log tail -n 10
30	Display entries from line 21 to line 30 of login.log	head -n 30 login.log tail -n 10
31	Display entries from line 11 to line 20 of access.log	head -n 20 access.log tail -n 10
32	Display entries from line 31 to line 40 of access.log	head -n 40 access.log tail -n 10
33	Display security events from line 21 to line 30	head -n 30 security.log tail -n 10
34	Display security events from line 51 to line 60	head -n 60 security.log tail -n 10
35	Display incident events from line 101 to line 110	head -n 110 incident.log tail -n 10
36	Display incident events from line 201 to line 220	head -n 220 incident.log tail -n 20
37	Display incident events from line 491 to line 500	head -n 500 incident.log tail -n 10

7. Filtering and Counting Together

No.	Task	Command
38	Take the first 20 entries of login.log and count the lines	head -n 20 login.log wc -l
39	Take the last 15 entries of access.log and count the lines	tail -n 15 access.log wc -l
40	Take the first 50 entries of security.log and count the words	head -n 50 security.log wc -w
41	Take the last 25 entries of security.log and count the lines	tail -n 25 security.log wc -l
42	Extract lines 21–30 from login.log and count the lines	head -n 30 login.log tail -n 10 wc -l
43	Extract lines 51–60 from access.log and count the lines	head -n 60 access.log tail -n 10 wc -l
44	Extract events 101–150 from incident.log and count the events	head -n 150 incident.log tail -n 50 wc -l
45	First 100 events , take the last 20 , then count lines	head -n 100 incident.log tail -n 20 wc -l